Advanced Biotechnology

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UNIVERSITY OF CAMBRIDGE

Lecture 2

Nucleic acids synthesis and assembly



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Lecture outline

1) Uses of synthetic DNA and RNA
2) Solid state synthesis of oligonucleotides
3) Assembly and enzymatic synthesis

Learning outcomes

1) Describe uses and commercial value of synthetic nucleic acids
2) Describe principles, chemistry and steps in solid state synthesis of oligonucleotides with phosphoramidite chemistry
3) Describe the capabilities and limitations of solid state nucleic acid synthesis
4) Describe methods for the enzymatic assembly of short oligonucleotides into longer constructs including:
    1) Polymerase cycling assembly
    2) Gibson assembly
5) Describe application of such methods to the synthesis of synthetic genes and genomes

Lecture outline

1) **Uses of synthetic DNA and RNA**
2) Solid state synthesis of oligonucleotides
3) Assembly and enzymatic synthesis

Synthetic DNA and RNA oligonucleotides

Short oligos of arbitrary sequence, used for:

- PCR primers
- Piece-wise synthesis of longer genes
- Guide RNA for CRISPR
- Small interfering RNA for therapeutic applications
- Biophysics and nanotechnology research
- ~$10B market size in 2025



Credit: Eurofins

Lecture outline

1) Uses of synthetic DNA and RNA
2) **Solid state synthesis of oligonucleotides**
3) Assembly and enzymatic synthesis

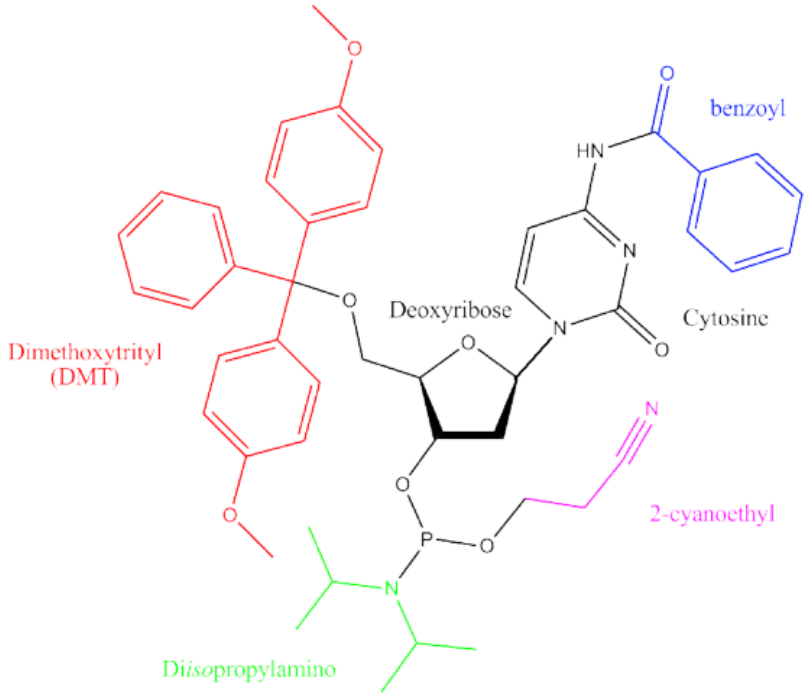
Solid state synthesis: overview

- Oligos are assembled one nucleotide at at time on a solid substrate (resin/beads)
- Enables complete automation
- Synthesis scale from ~n mol to ~m mol (~µ mol is typical)
- Enables inclusion of non-natural nucleotides and chemical modifications
- Used for both RNA and DNA
- Performed in organic solvents (e.g. acetonitrile)
- Yield decreases with product length, so limited to oligos with < 200 nt



An oligo synthesiser
Credit: CSbio

Solid state synthesis: phosphoramidite chemistry

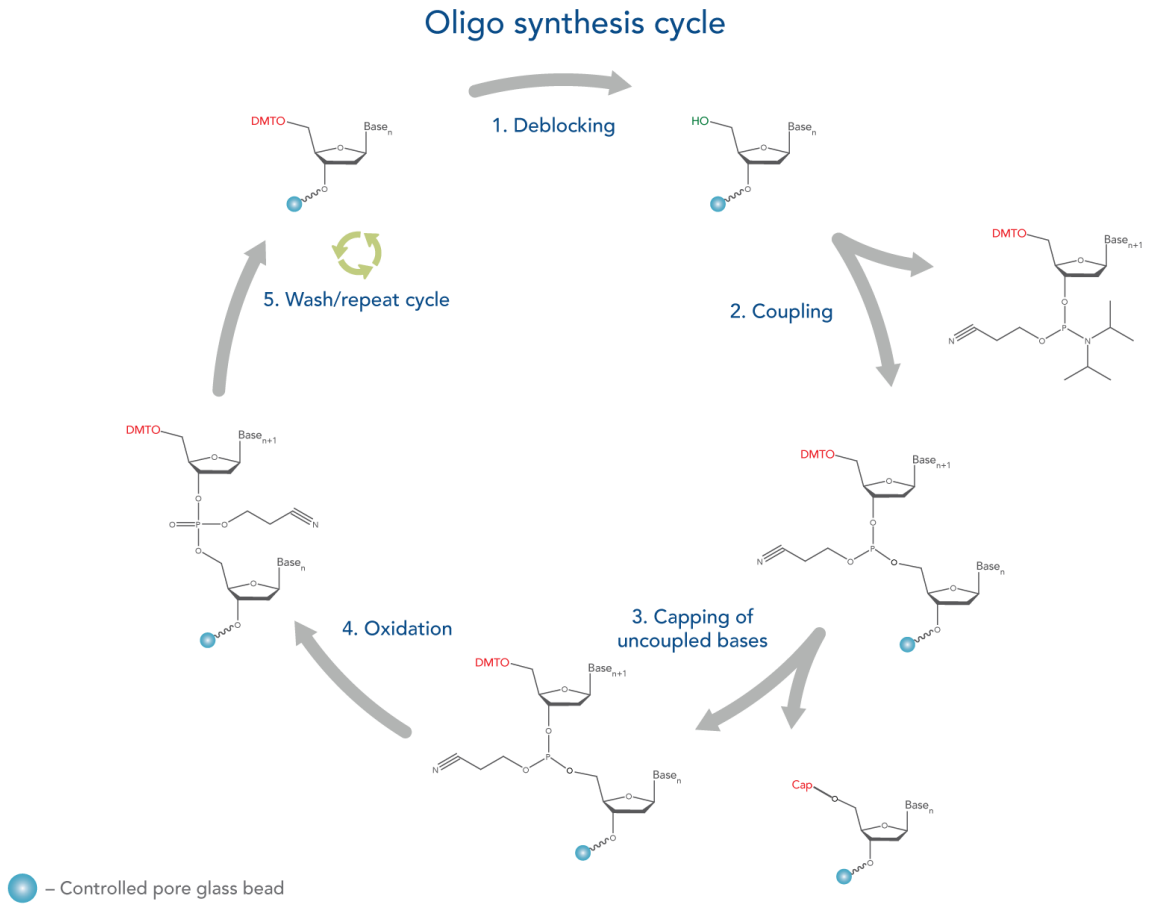


A cytosine phosphoramidite

Credit: Integrated DNA Technologies (IDT)

- Phosphoramidite monomers are nucleotides in which functional groups are protected to prevent unwanted reactions
- Protective groups can vary, but the DMT group protecting the 5' OH is always used
- Phosphoramidites exist for all deoxyribonucleotides and ribonucleotides (both for RNA and DNA)

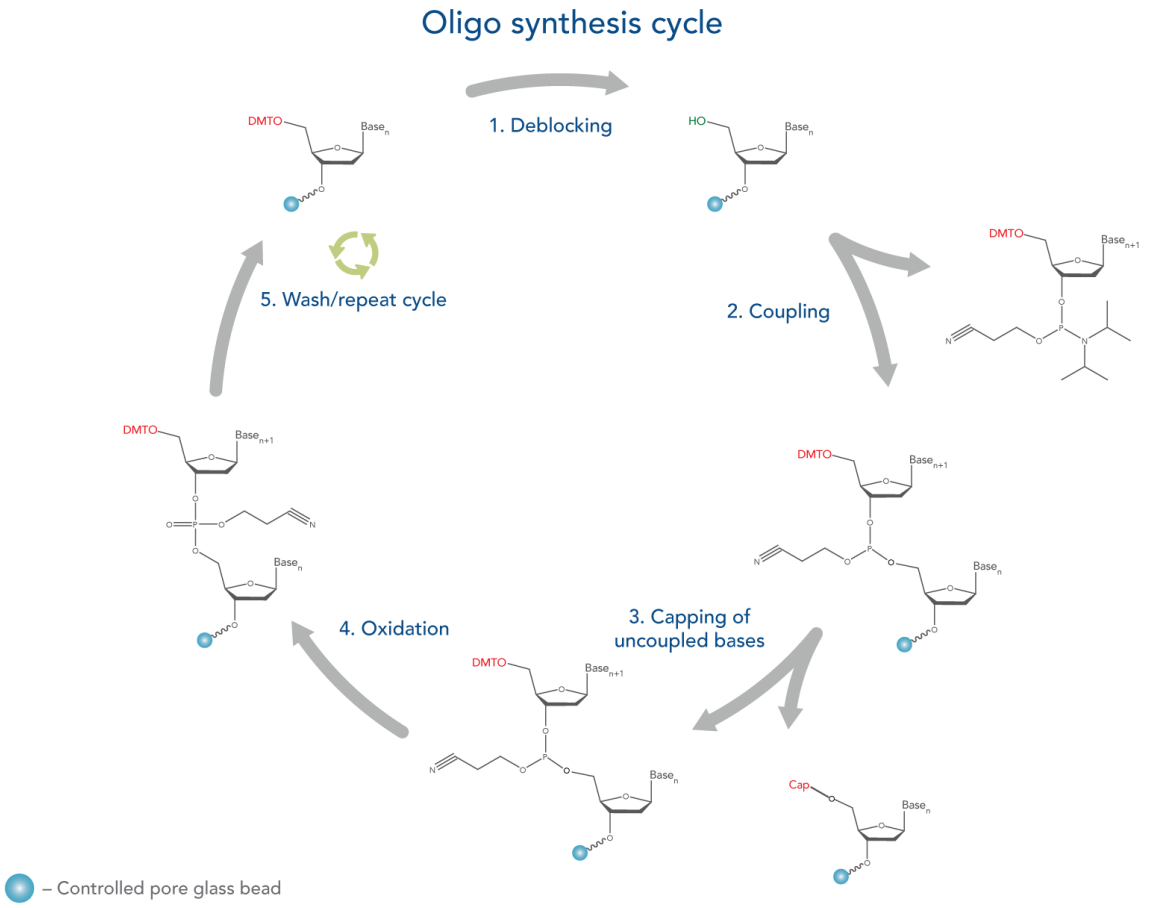
Solid state synthesis: cycle



Credit: Integrated DNA Technologies (IDT)

- Star with one protected monomer coupled to a solid substrate (bead, resin) through a cleavable linker
- **Deblocking:** Deprotect the 5' OH group by removing DMT
- **Coupling:** Introduce the next monomer, which reacts with the unprotected monomer on the substrate
- An (unstable) phosphite triester linkage bond is formed between the two monomers

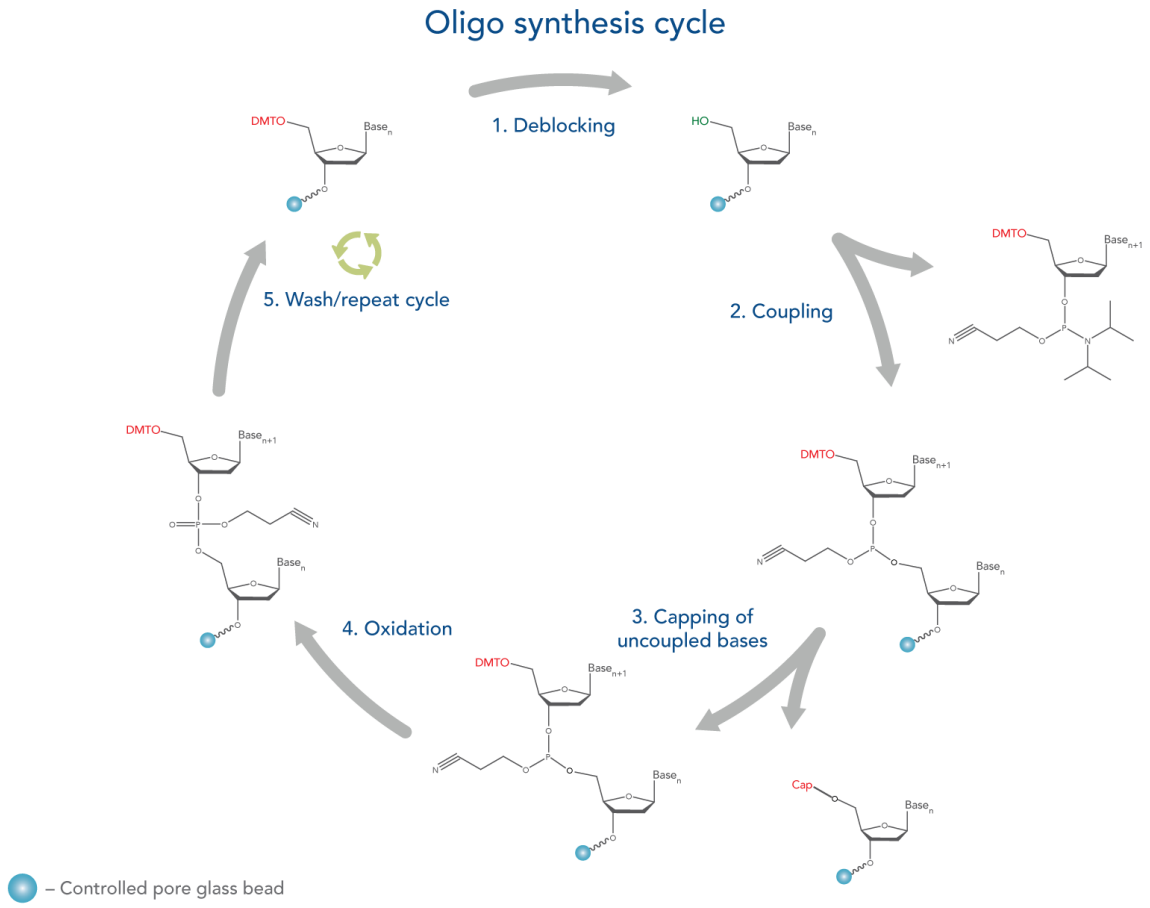
Solid state synthesis: cycle



Credit: Integrated DNA Technologies (IDT)

- The coupling efficiency is never 100% (commercial providers claim >99%)
- The free 5' OH group on unreacted monomers is therefore **capped** to prevent unwanted reactions with the subsequent monomer, which would result in undesirable **deletion defects**
- **Capping of uncoupled bases** can be achieved with different chemistries
- These capped oligos will no longer be extended in subsequent steps and will therefore form **truncated products**, which can be removed through purification (much easier to remove than deletion defects)

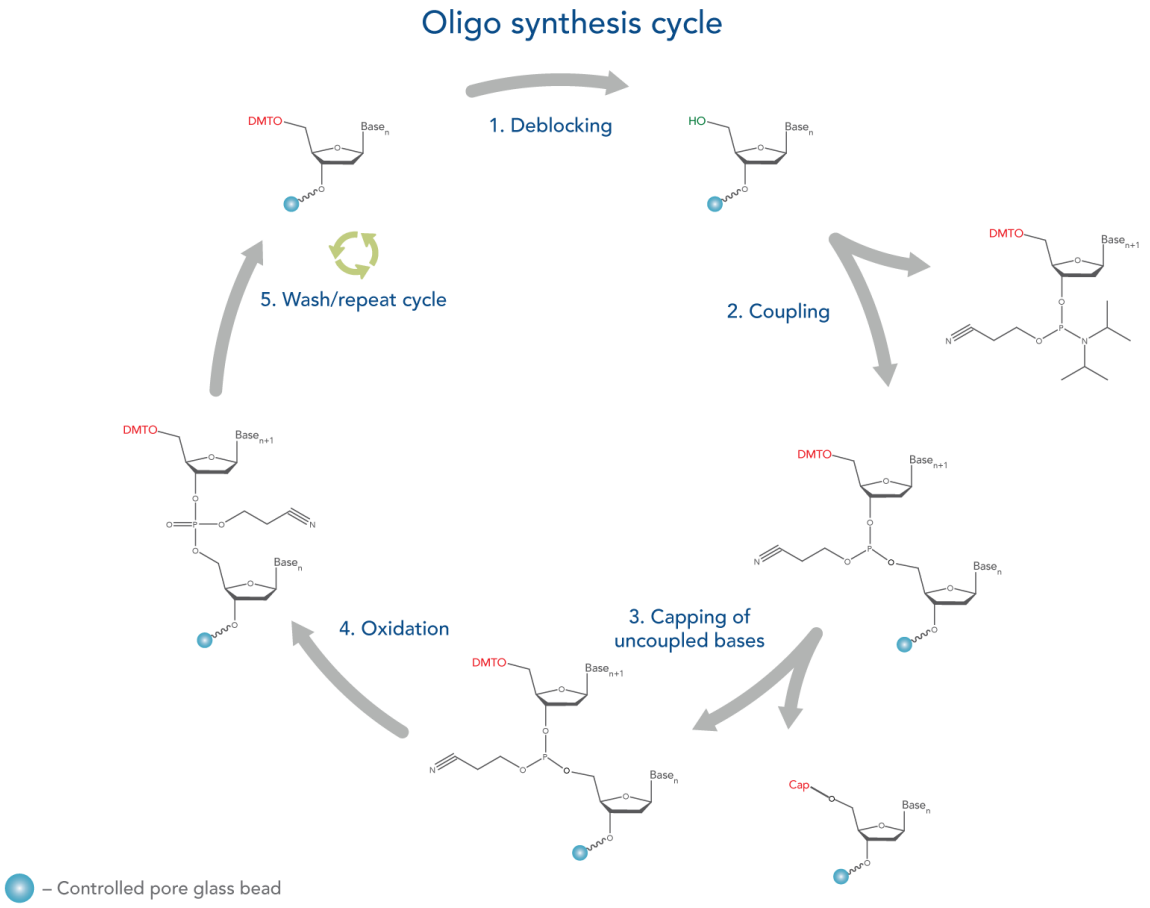
Solid state synthesis: cycle



Credit: Integrated DNA Technologies (IDT)

- The unstable phosphite triester bond is oxidised to form a stable phosphate triester bond
- Unreacted compounds are washed away
- The cycle is repeated to add the next monomer, for the desired number of steps
- After the final cycle, the resulting oligos are de-protected to remove all residue capping groups and detached from the substrate
- **Note that synthesis happens always in the 3' to 5' direction**

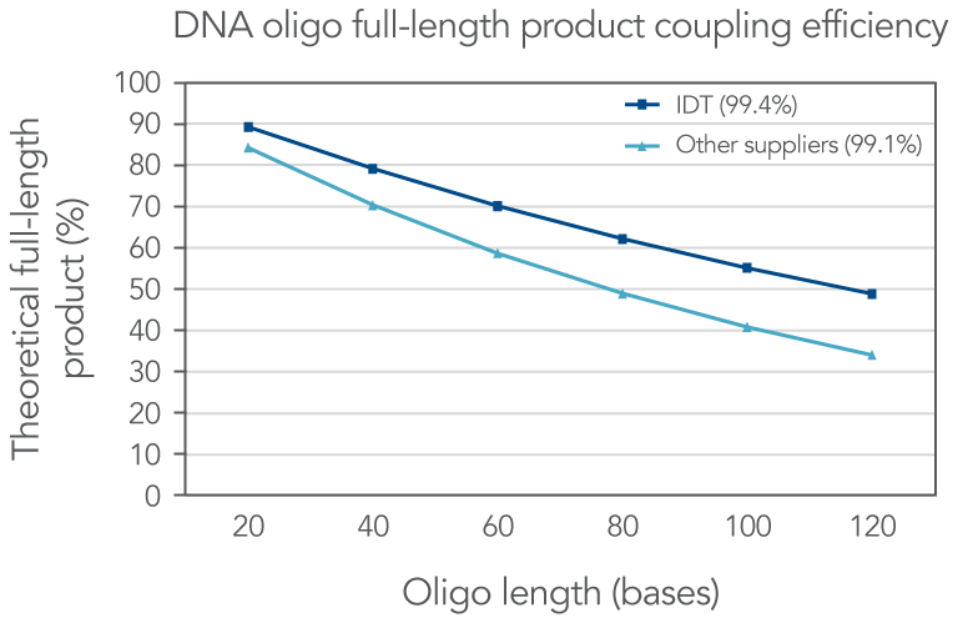
Solid state synthesis: cycle

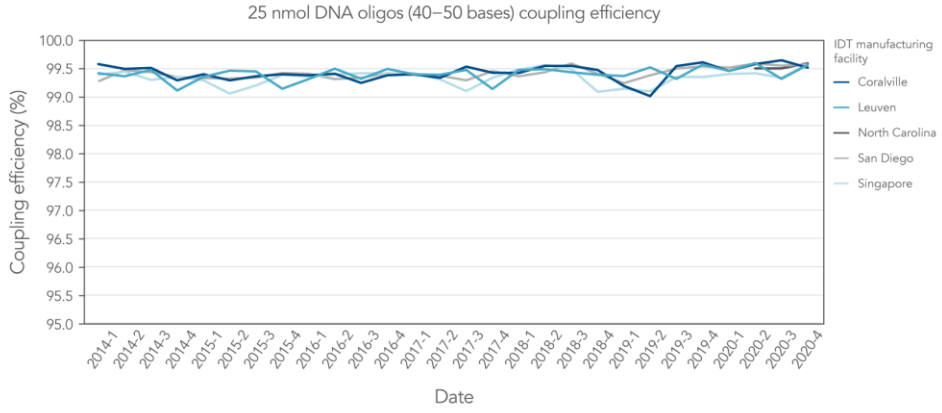


Credit: Integrated DNA Technologies (IDT)

- The detached oligos are finally purified by **desalting**, to remove all unreacted compounds and organic salts present
- Desalting can be conducted, e.g., with size exclusion chromatography, precipitation or diafiltration

Solid state synthesis: yield

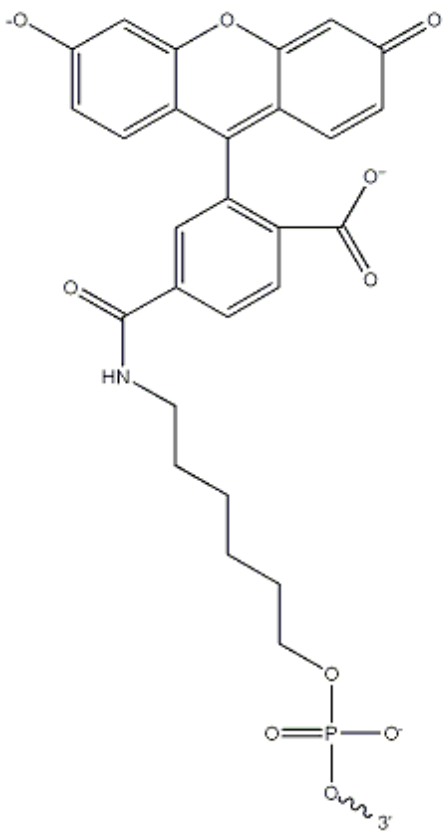




Credit: Integrated DNA Technologies (IDT)

- Because coupling efficiency is <100%, the fraction of truncated products increases with oligo length
- For ~60-80 nt, the yield of full-length products can drop to <50%
- Standard desalting **does not** remove truncated products
- Alternative purification is thus required to remove truncated products, namely
  - **High Performance Liquid Chromatography** (HPLC, more on this later in the course)
  - **Polyacrylamide Gel Electrophoresis** (PAGE, you have already studied this)

Solid state synthesis: chemical modifications

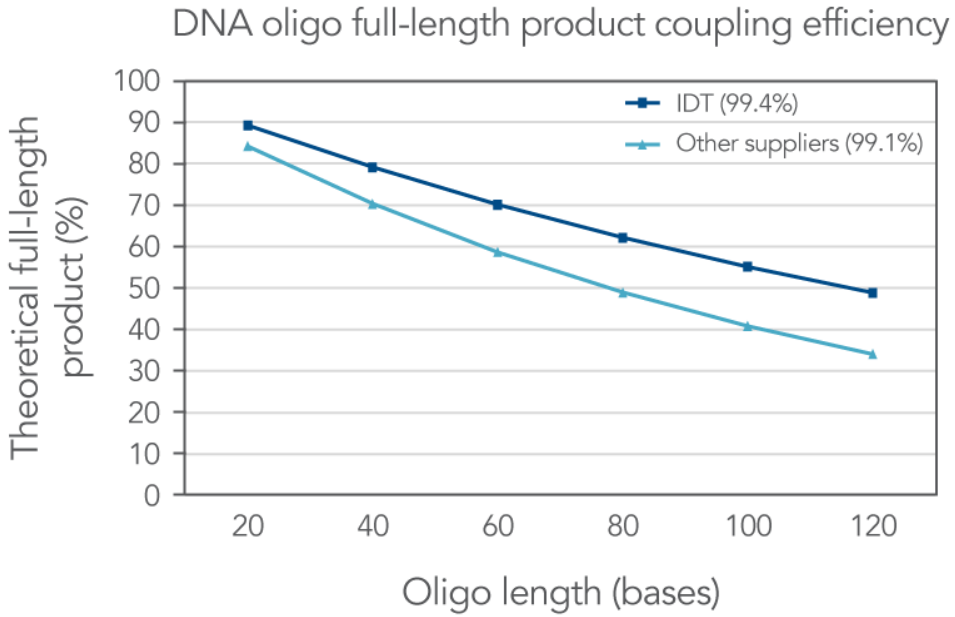


A 3' fluorescein modification

Credit: Integrated DNA Technologies (IDT)

- Phosphoramidite synthesis enables incorporation of arbitrary chemical modifications, either at the 3' end, the 5' end or internally
- Possible modifications include
  - **Fluorescent molecules or quenchers**
  - **Reactive linkers** to connect the oligos to other materials/molecules (e.g. click chemistry linkers)
  - **Spacers**
  - **Modified nucleotides** (e.g. methylated, locked)

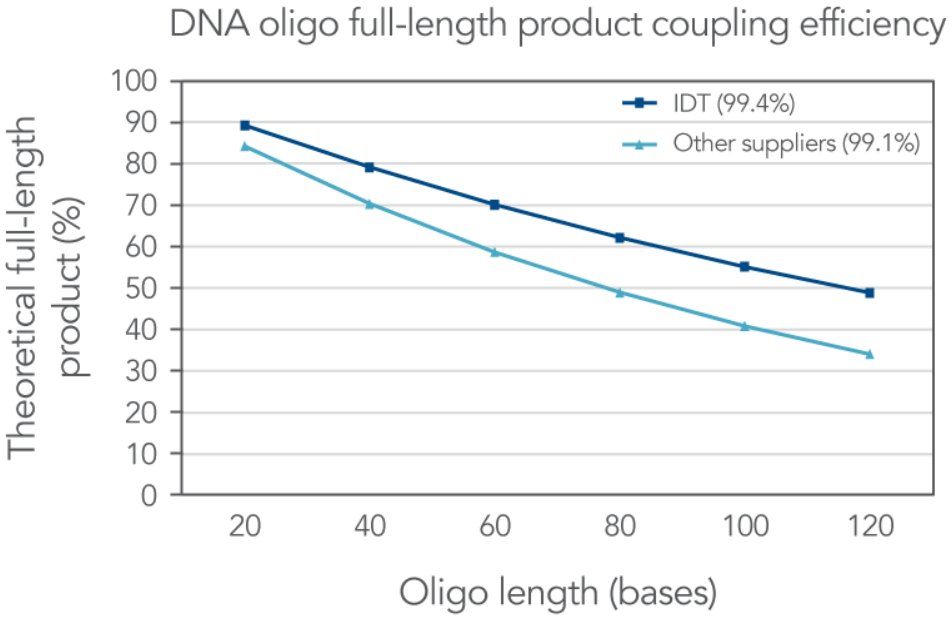
Exercise



Integrated DNA Technologies claims a coupling efficiency of 99.4% in their oligo synthesis process, as opposed to 99.1% of "other suppliers".

1. Derive an expression for the yield of full-length products vs oligo length
2. Estimate the yield for a 20, 60 and 200 nt oligos for IDT and "other suppliers"

Solution



- If P is the coupling probability, the yield for an n-mer is simply $Y(n) = P^{n-1}$
- IDT claims $P = 0.994$, hence

$$Y(20) = 0.994^{19} = 0.892$$
$$Y(60) = 0.994^{59} = 0.701$$
$$Y(200) = 0.994^{199} = 0.302$$

- For "other suppliers"

$$Y(20) = 0.991^{19} = 0.842$$
$$Y(60) = 0.991^{59} = 0.587$$
$$Y(200) = 0.991^{199} = 0.165$$

Lecture outline

1) Uses of synthetic DNA and RNA
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3) **Assembly and enzymatic synthesis**

From oligos to synthetic genes?

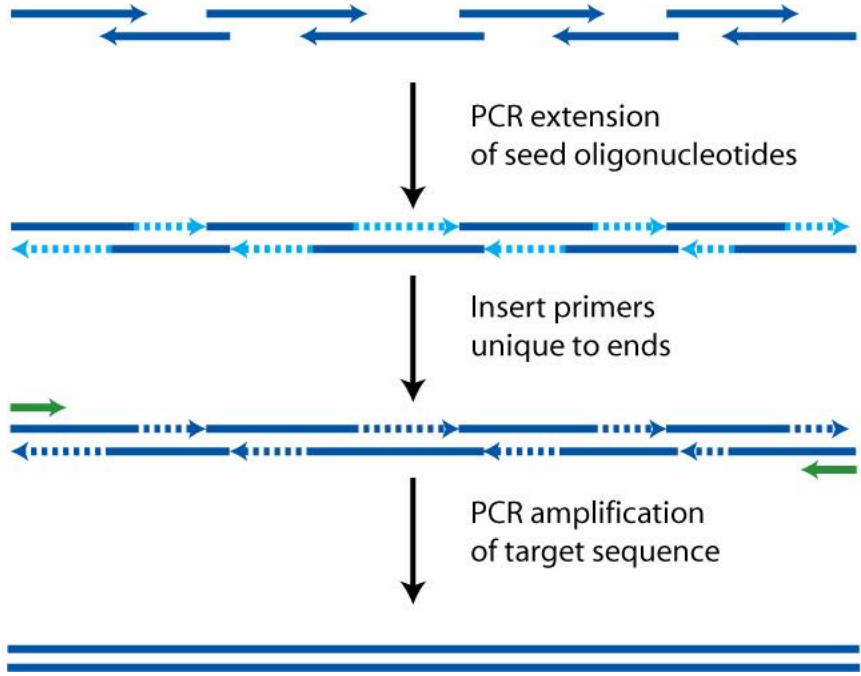
- For many biotechnology and synthetic biology applications one can start off from existing (natural) genes, and then engineer them by introducing mutations etc (you have learned this in biotech fundamentals)
- However, many modern applications require constructing genes completely de novo
- How do we go from short synthetic oligos to full scale genes?



Credit: Eurofins

Polymerase cycling assembly

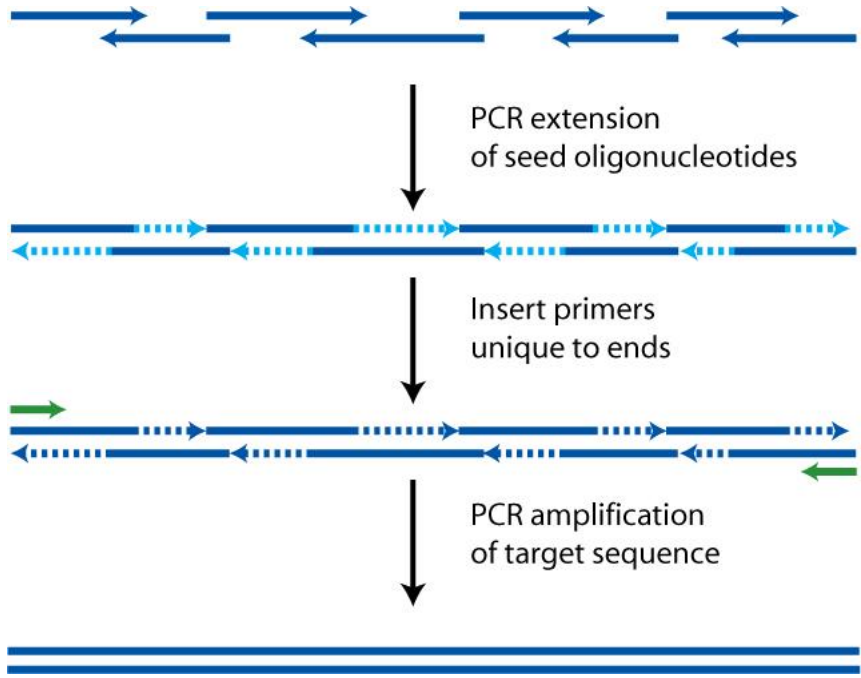
- The simplest method to connect short synthetic oligonucleotides into long double-stranded segments of several kb (kilo bases)
- Start off with DNA oligonucleotides (~50-60 nt) that share complementary overlap regions (~20 nt)
- Mix all oligos in equimolar concentration such that they can (potentially) anneal to form the target, full-length chain



Original reference: Stemmer et al, Gene 164(1), 49-53, 1995. **Please read!**

Polymerase cycling assembly

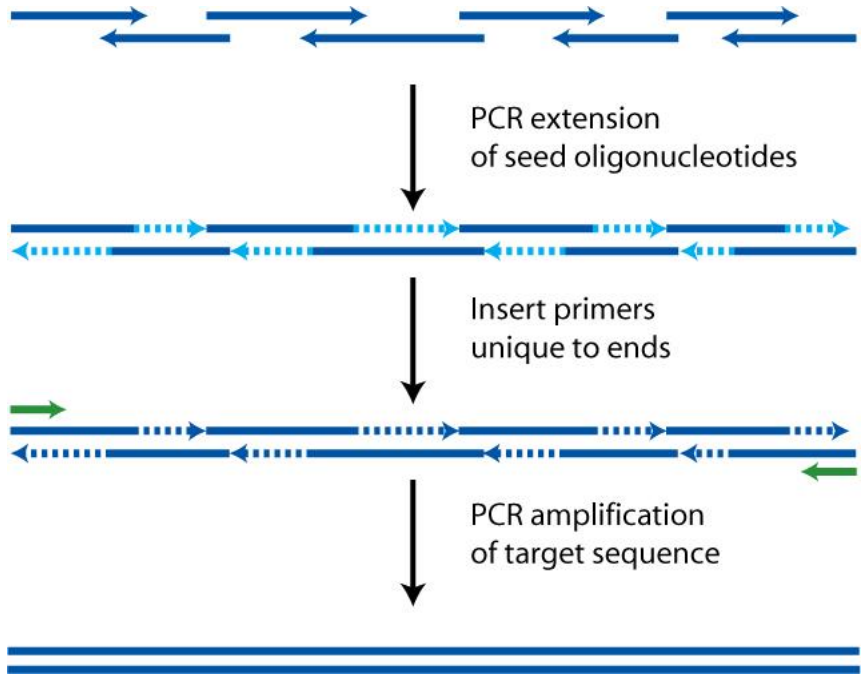
- Like PCR primers, it is required that the overlap regions between oligos have similar melting temperature and that the oligos are unlikely to form strong secondary structure
- Run ~30 PCR cycles with the mixture of oligos **without additional primers**
- During this PCR extension phase hybridised oligos will be extended the polymerase



Original reference: Stemmer et al, Gene 164(1), 49-53, 1995. **Please read!**

Polymerase cycling assembly

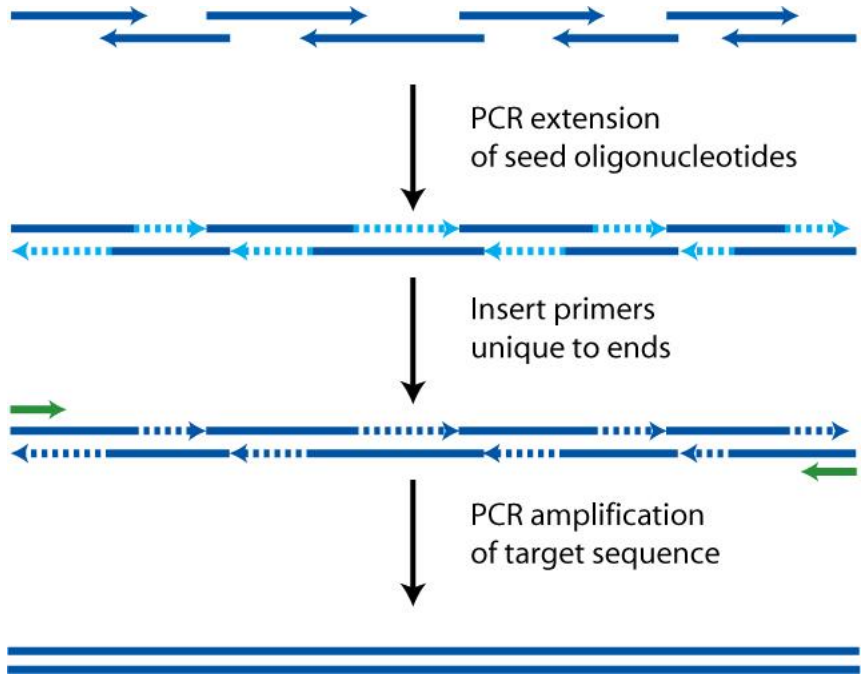
- Constructs of different lengths will be formed, depending on which oligos happened to be hybridised
- After ~30 cycles, there is a good chance that some of the full-length products are produced
- Add primers that bind the ends of the full-length products and run additional ~20 PCR cycles to selectively amplify the full-length product



Original reference: Stemmer et al, Gene 164(1), 49-53, 1995. **Please read!**

Polymerase cycling assembly

- Potentially purify (e.g. with gel electrophoresis) to remove shorter products
- Can produce gene fragments up to a few kb under optimal conditions
- Creating longer constructs (e.g. synthetic genomes) requires alternative methos



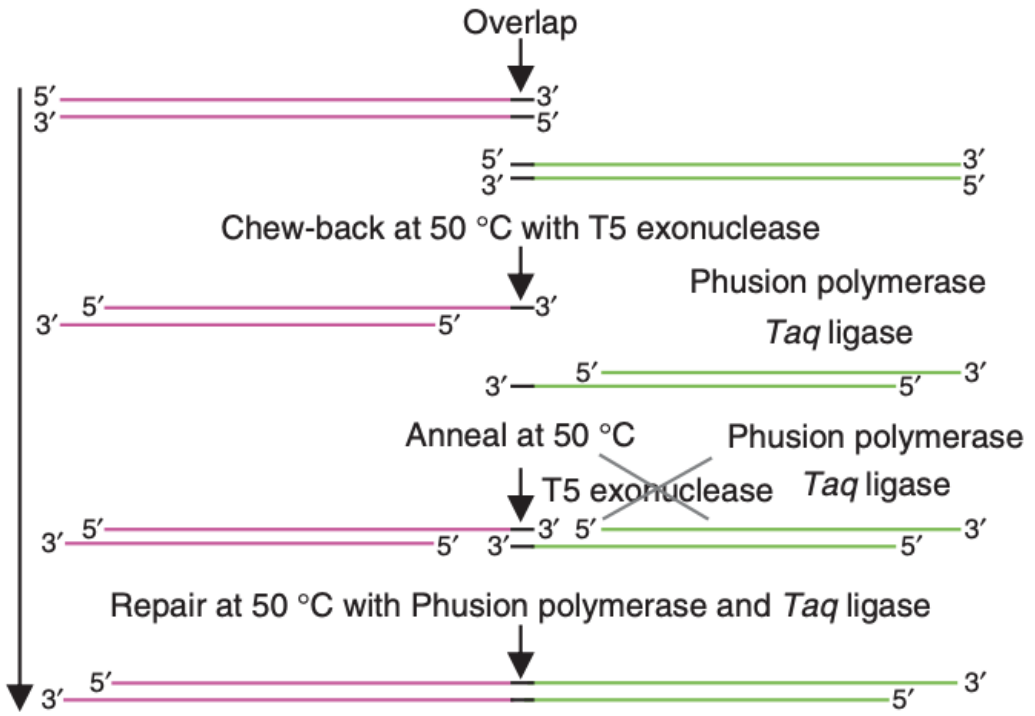
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Gibson assembly



Gibson assembly

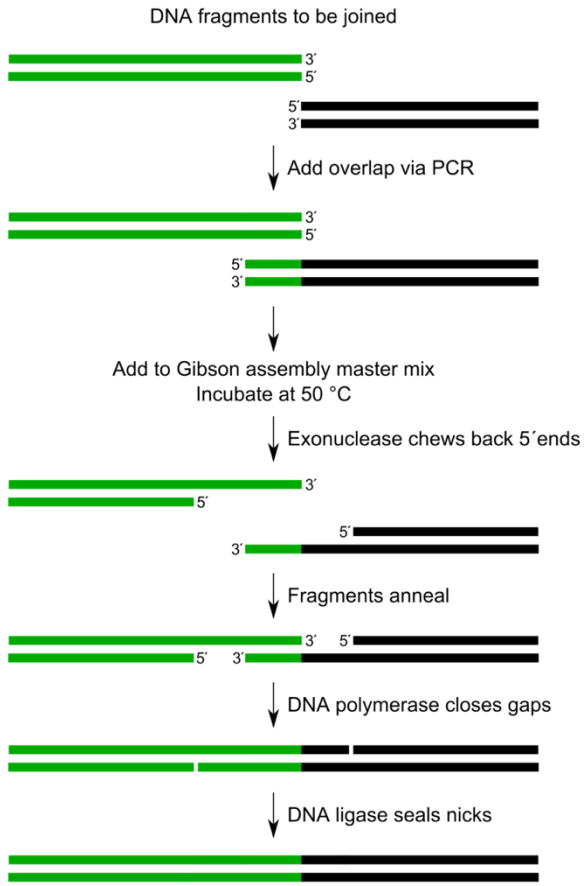
- Start with two double-stranded gene fragments that share overlapping regions (~50 bp) at one end (black in the diagram)
- Overlapping regions can be added with PCR primers
- T5 exonuclease chews back the 5' ends, creating complementary overhangs that allow the fragments to connect



Original reference: Gibson et al, Nature Methods 6, 343–345, 2009. **Please read!**

Gibson assembly

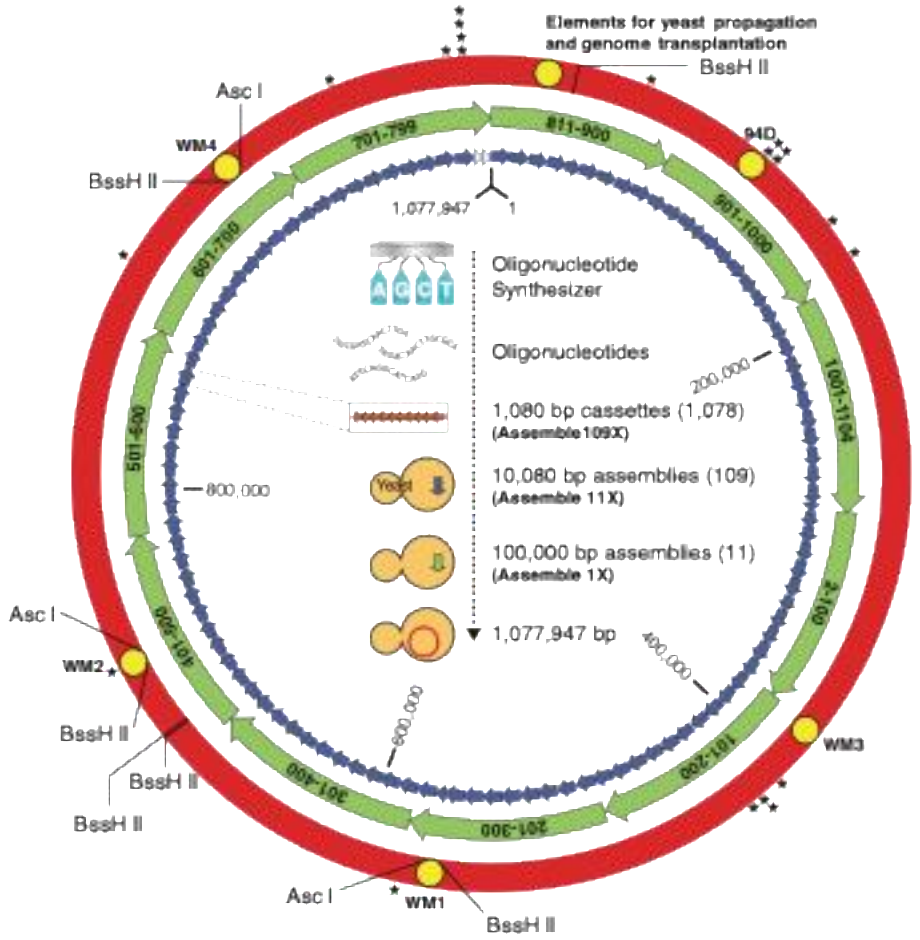
- Phusion polymerase extends back the 5' ends to reconstruct the excess digested regions
- Taq ligase then repairs the nicks to produce the target construct
- Can be conducted in a one pot, isothermal reaction at 50 °C or in a two-step reaction in which first one adds the exonuclease and then the polymerase and ligase
- The two-step reaction is preferred for longer constructs



Original reference: Gibson et al, Nature Methods 6, 343–345, 2009. **Please read!**

Synthetic genomes: *M. mycoides*

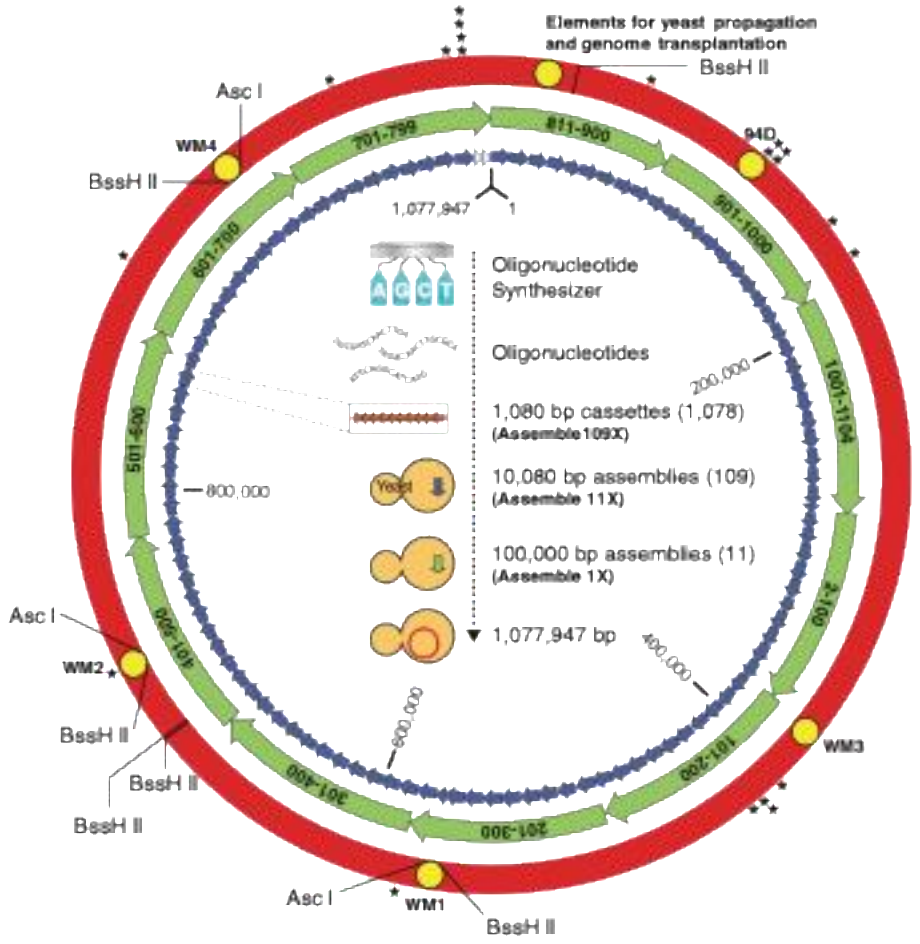
- Can incorporate very long constructs (up to 100 kb) and create entire synthetic genomes
- In 2010 Gibson et al (same guy) form the Craig Venter Institute (JCVI) synthesised an entire synthetic genome using these techniques
- They started from sequencing (next lecture!) information on the genome of *Mycoplasma mycoides* (a very simple bacterium with a short genome)



Original reference: Gibson et al, Science 329, 53-56, 2010. **Please read!**

Synthetic genomes: *M. mycoides*

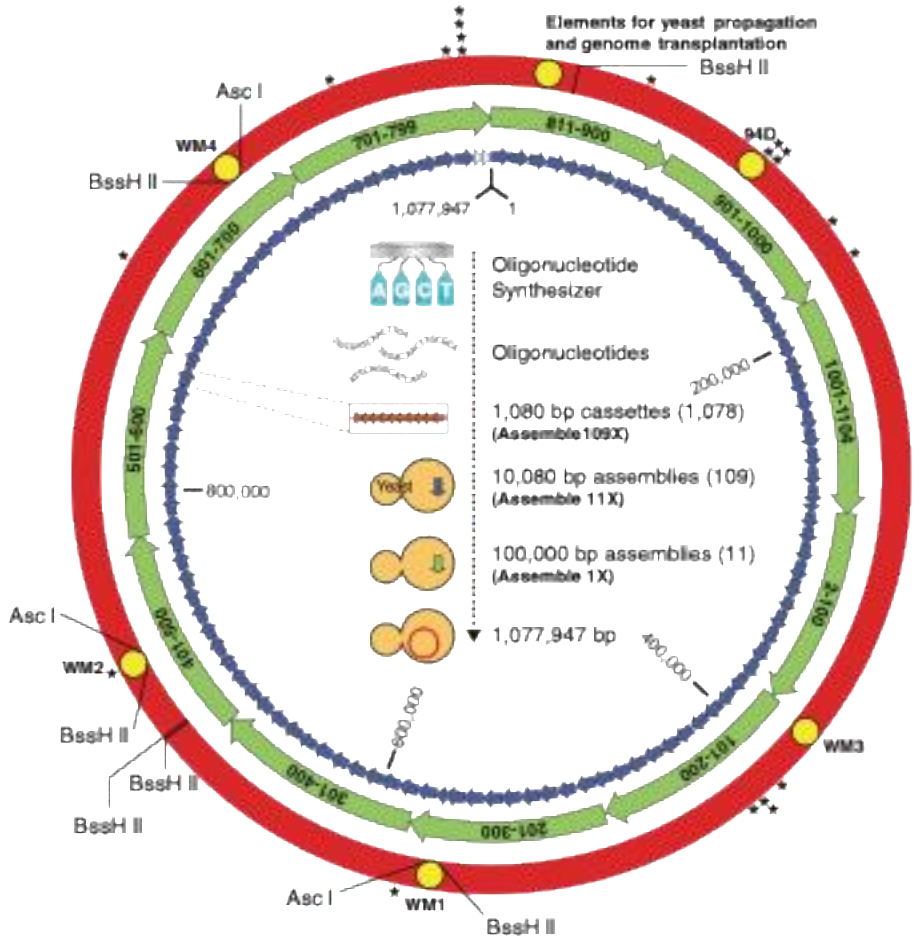
- They then proceeded to assemble a fully synthetic *M. mycoides* genomes hierarchically
- Started from DNA oligonucleotides produced with solid state synthesis and assembled larger and and larger fragments
- For the largest fragments, **in vivo recombination in yeast was used**



Original reference: Gibson et al, Science 329, 53-56, 2010. **Please read!**

Synthetic genomes: *M. mycoides*

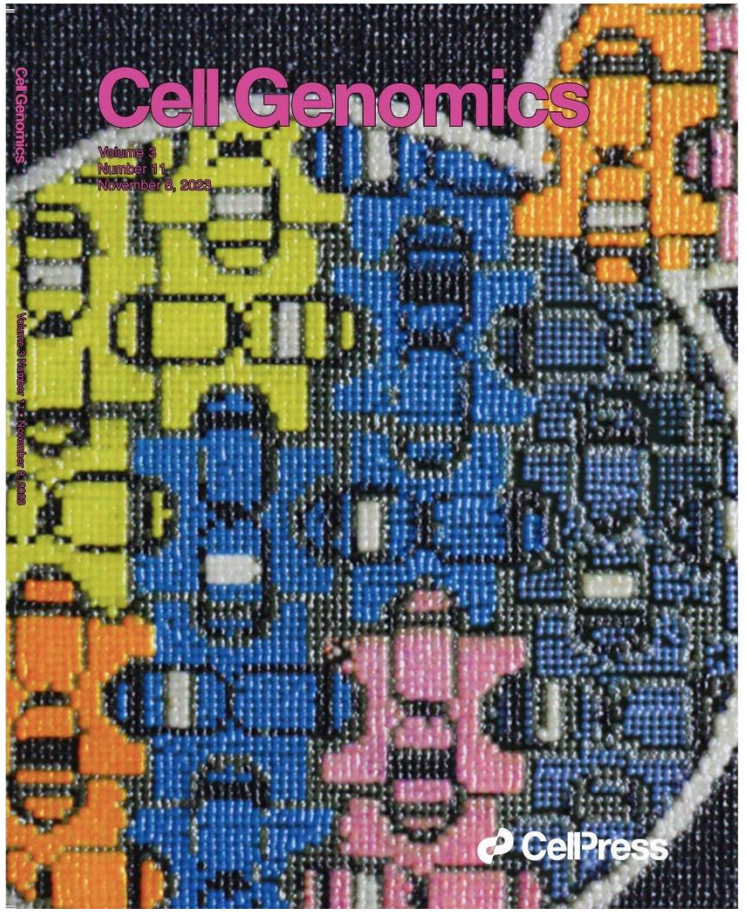
- The synthetic DNA was inserted into a recipient cell *M. capricolum* to create a strain of *M. mycoides* with entirely synthetic genome, dubbed JCVI-syn1.0
- The JCVI team proceeded to produce subsequent iterations trying to minimise the genome and create **a minimal bacterial cell**. Current version is JCVI-syn3.0 (more on this later)



Original reference: Gibson et al, Science 329, 53-56, 2010. **Please read!**

Synthetic genomes: Synthetic Yeast Project (*Saccharomyces cerevisiae* 2.0, Sc2.0)

- An international consortium attempting to produce **the first completely synthetic yeast (eukaryotic) genome**
- According to syntheticyeast.github.io, the team has synthesised 14 of the 16 chromosomes required



Synthetic genomes: Synthetic Human Geomne project (SynHG)

- Recently (June 2025) funded project by the Wellcome Trust: ~£10M to synthesise a complete human genome
- Lead by Jason Chin (MRC Laboratory of Molecular Biology, just moved to Oxford)

https://www.bbc.co.uk/news/articles/c6256wpn97ro